Project: Worked examples
Title: Thermal expansion of a restrained bar
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

A bar heated or cooled between two restraints: its free movement, the part closed by a gap before the restraint engages, the restraint force for a rigid or an elastic restraint, the axial stress that results and the movement that actually occurs, with the stress against an allowable.
$\delta_{free} = \alpha \cdot L \cdot \Delta T$. $\delta_e = \text{sign}(\delta_{free}) \cdot \max(|\delta_{free}| - g, 0)$. $k_{bar} = E \cdot A / L$. $F = \delta_e \cdot k_{bar}$ (rigid) or $\delta_e / (1/k_{bar} + 1/k_s)$ (elastic restraint). $\sigma = -F/A$ (compressive on heating). $\delta = \delta_{free} - F/k_{bar}$.
Ref: Roark's Formulas for Stress and Strain Ch. 15 — Thermal stresses
Ref: Shigley's Mechanical Engineering Design §3-17 — Thermal stresses and strains

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L	Bar length	2	m	drawing: DRG-TX-501 rev A	0.01 ... 1000 m
A	Cross-section area	500	mm^2	drawing: DRG-TX-501 rev A	1 ... 1000000 mm^2
E	Modulus of elasticity	200	GPa	drawing: DRG-TX-501 rev A	1 ... 500 GPa
alpha	Coefficient of thermal expansion	0.000012	1/K	drawing: DRG-TX-501 rev A	1e-7 ... 0.0001 1/K
dT	Temperature change (positive heating)	60	K	spec: LOAD-SCHEDULE-01	-300 ... 1000 K
g	Gap closed before the restraint engages	0.5	mm	drawing: DRG-TX-501 rev A	0 ... 1000 mm
sigma_allow	Allowable stress	150	MPa	spec: SPEC-MECH-03 design criteria	1 ... 2000 MPa

3. Assumptions

A1. The expansion coefficient is constant over the temperature change and the bar stays elastic. (Linear thermoelasticity.)
A2. The bar is uniformly heated and does not buckle under the compressive force. (Check buckling separately (struct.column-buckling).)

4. Working

`[8] delta_free = alpha * L * dT` — Shigley §3-17
`= 0.000012 1/K × 2 m × 60 K`
`= 1.44 mm`
Free thermal movement (positive on heating)
`[9] delta_e = delta_free - g`
`= 1.44 mm - 0.5 mm`
`= 0.94 mm`
Movement after the gap closes (engaged movement)

[10] k_bar = E * A / L
= 200 GPa × 500 mm² / 2 m
= 50 kN/mm
Axial stiffness of the bar
[11] F = delta_e * k_bar
= 0.94 mm × 50 kN/mm
= 47000 N
Restraint force (rigid restraint)
Roark Ch. 15
[12] sigma = -F / A
= -47000 N / 500 mm²
= -94 MPa
Axial stress (negative compressive)
[13] delta = delta_free - F / k_bar
= 1.44 mm - 47000 N / 50 kN/mm
= 0.5 mm
Actual movement
[14] u = abs(sigma) / sigma_allow
= abs(-94 MPa) / 150 MPa
= 0.62667
Stress utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
delta_free	1.44 mm	Free movement
delta_e	0.94 mm	Engaged movement
k_bar	50 kN/mm	Bar stiffness
F	47000 N	Restraint force
sigma	-94 MPa	Axial stress
delta	0.5 mm	Actual movement
u	0.62667	Stress utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Thermal stress against allowable	u = 0.62667	u ≤ 1	0.627	

7. Applicability

No warnings.

8. Verification

Template mech.thermal-expansion@1.0.0 Content hash 63e435d7263ca8d1 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0